# Ten-Layer Reset-Route Review

## A Typed Native–Directional Comparison and the Two-Gate Open Frontier

Prime Dynamics Theory Program

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**Abstract**

RH-151–RH-158 replaced an unstable recursive packet transport by independently certified spectral resets and then developed two different output objects: a native compressed support and a projected directional cross. We review the result as a typed ten-gate proof system rather than treating every positive number as interchangeable.

The main new theorem compares the types. If a positive operator has block form

$$R = \begin{pmatrix} A & B^* \\ B & C \end{pmatrix} \succeq 0, \qquad A = PRP, \quad B = (I-P)RP,$$

then $B^*B \preceq \|C\|A$. Hence a fourth directional-cross lower $\gamma$ and complement upper $U$ imply $\lambda_4(A) \geq \gamma^2/U$. The reverse implication is false: a uniformly positive native compression can coexist with $B = 0$. Thus directional data can dominate native positivity under an energy upper, but native support cannot be silently substituted for a directional cross.

The audited route lattice has seven finite-certified gates, one exact obstruction, and two open gates. Native reset and adaptive lag are the two finite surviving seeds. Contemporaneous directional reset is exactly obstructed, while the current branch-free recursive bound is finitely noninformative. None is program-complete because uniform all-level endpoint laws and typed downstream assembly remain open. All eight input archives, 96 publication artifacts, 319 hash assertions, and 16 acyclic internal dependency edges are independently reverified. No Stage A, Hilbert–Polya, zero-identification, or Riemann-Hypothesis conclusion is made.

# 1 Why the route must be typed

The reset block produced two successful finite constructions. RH-156's native object depends on positive compressions $PRP$ and $PTP$. RH-158's directional object depends on the cross action $(I-P)RP$. RH-157 showed why their distinction matters: for a contemporaneous spectral projector of $M = R + T$,

$$(I-P)RP = -(I-P)TP.$$

Positive native energy therefore gives no current-cross lower. Lag restores the cross by exposing recent innovations, but it also changes the packet time. The present review records which route proves which type and identifies the first genuinely common frontier.

## 2 Directional data imply native data in one direction

We first isolate a general positive-block theorem.

**Theorem 1** (Cross-to-compression implication). *Let $R \succeq 0$ and let $P$ be an orthogonal projector. Relative to $P\mathcal{H} \oplus (I-P)\mathcal{H}$, write*

$$R = \begin{pmatrix} A & B^* \\ B & C \end{pmatrix}.$$

*Then*

$$B^*B \preceq \|C\|A.$$

*In particular, if $\|C\| \le U$ and $\sigma_j(B) \ge \gamma_j$, then*

$$\lambda_j(A) \ge \frac{\gamma_j^2}{U}$$

*whenever $U > 0$.*

*Proof.* For $\varepsilon > 0$, positivity and the Schur complement give $B^*(C+\varepsilon I)^{-1}B \preceq A$. Since $C + \varepsilon I \preceq (\|C\| + \varepsilon)I$, inversion reverses order and

$$B^*B \preceq (\|C\| + \varepsilon)A.$$

Let $\varepsilon \downarrow 0$. Eigenvalue monotonicity applied to $B^*B \preceq UA$ gives the second assertion [Bhatia, 1997, Horn and Johnson, 2013]. □

**Proposition 2** (Strict reverse nonimplication). *No positive lower on any eigenvalue of $A = PRP$ implies a positive lower on any singular value of $B = (I-P)RP$, even for $R \succ 0$.*

*Proof.* Take $R = A \oplus C$ with $A \succeq \ell I$ and $C \succ 0$. Then $B = 0$ while the native compression has lower $\ell > 0$. □

For the depth-five geometric recent Gram, trace-one snapshots give the dynamics-free complement upper

$$U_5 \le \sum_{j=0}^{4} \eta^j = 1.0019569472.$$

RH-158's common selected fourth-cross lower $3.18077 \times 10^{-16}$ therefore implies a same-lag-packet native fourth eigenvalue lower $1.00975 \times 10^{-31}$. This is positive but extremely small. It is not RH-156's current-packet support floor $3.26204 \times 10^{-8}$, whose overlap and tail factors are different. Theorem 1 compares logical strength, not numerical interchangeability.

## 3 The ten gates

Table 1 separates finite facts, obstructions, and open claims. "Finite" always refers to the five frozen scales and their archived outward balls; it is not shorthand for an asymptotic theorem.

The correlation entry is essential. Simultaneously transporting a pair by one inverse congruence preserves its Loewner tail ratio exactly; replacing that correlated image by independent balls loses positivity at 52 of 120 transitions. This is an information-loss obstruction, not evidence that the exact pair is indefinite.

| gate | object | evidence | status / sharp consequence |
|---|---|---|---|
| 1 | direct packet localization | RH-151 | finite: 130/130 reset balls |
| 2 | transition coherence | RH-152 | finite: 120/120 invertible |
| 3 | correlated congruence | RH-153 | finite: 120/120; independent only 68 |
| 4 | delayed terminal half | RH-154 | finite: 62 transitions, overlap $\geq 0.06664$ |
| 5 | native recent/tail pair | RH-155 | finite: 130/130 subunit |
| 6 | native support floor | RH-156 | finite: 120/120, floor $3.2620 \times 10^{-8}$ |
| 7 | contemporaneous cross | RH-157 | obstruction: 50 exact zeros; 54/80 active |
| 8 | adaptive lagged cross | RH-158 | finite: 120/120 for lag $\leq 8$ |
| 9 | uniform endpoint laws | — | open: no all-level gap/overlap/tail/lag law |
| 10 | typed downstream assembly | — | open: cap/quotient/determinant not recomposed |

Table 1: The ten reset-route gates after RH-158.

# 4 Route-lattice classification

We distinguish a *finite seed destination* from program closure. A route is finite-closed at a destination if every required gate is proved or certified; it is rejected if a required gate has an obstruction; otherwise it is open.

**Proposition 3** (Audited route classification)**.** *At the finite-seed destination, exactly two routes survive:*

1. *the native route through gates* $1, 2, 3, 5, 6$*;*
2. *the lagged directional route through gates* $1, 2, 8$*.*

*The recursive route is rejected by its current branch-free packet bound, and the contemporaneous directional route is rejected by gate 7. At full program closure, no route is closed because every survivor still requires gates 9 and 10.*

*Proof.* RH-151 certifies all direct resets but its universal branch-free scalar transport remains informative at only 11 of 130 snapshots. Gates $1, 2, 3, 5, 6$ are all finite-certified, so the native seed closes. The exact RH-157 cancellation is a required obstruction for the contemporaneous cross. Gates $1, 2, 8$ are finite-certified, so the lagged cross seed closes. Finally, the audit records gates 9 and 10 as open for both survivors. This is the truth-table evaluation of the stated required sets. □

Proposition 3 does not rank the two survivors without a target type. If a later theorem accepts native compression support, the native route avoids an unnecessary cross hypothesis. If it requires four cross images, Proposition 2 forbids replacing that input by native positivity, and the lagged route is the only surviving finite choice.

# 5 Archive and numerical audit

The review rehashes every declared publication artifact, result, local source, and external input for RH-151–RH-158. All 319 assertions match. The 96 publication artifacts are intact, and the 16 internal dependency edges form a directed acyclic graph. This also checks that later conclusions use archived earlier inputs rather than circularly importing their own outputs.

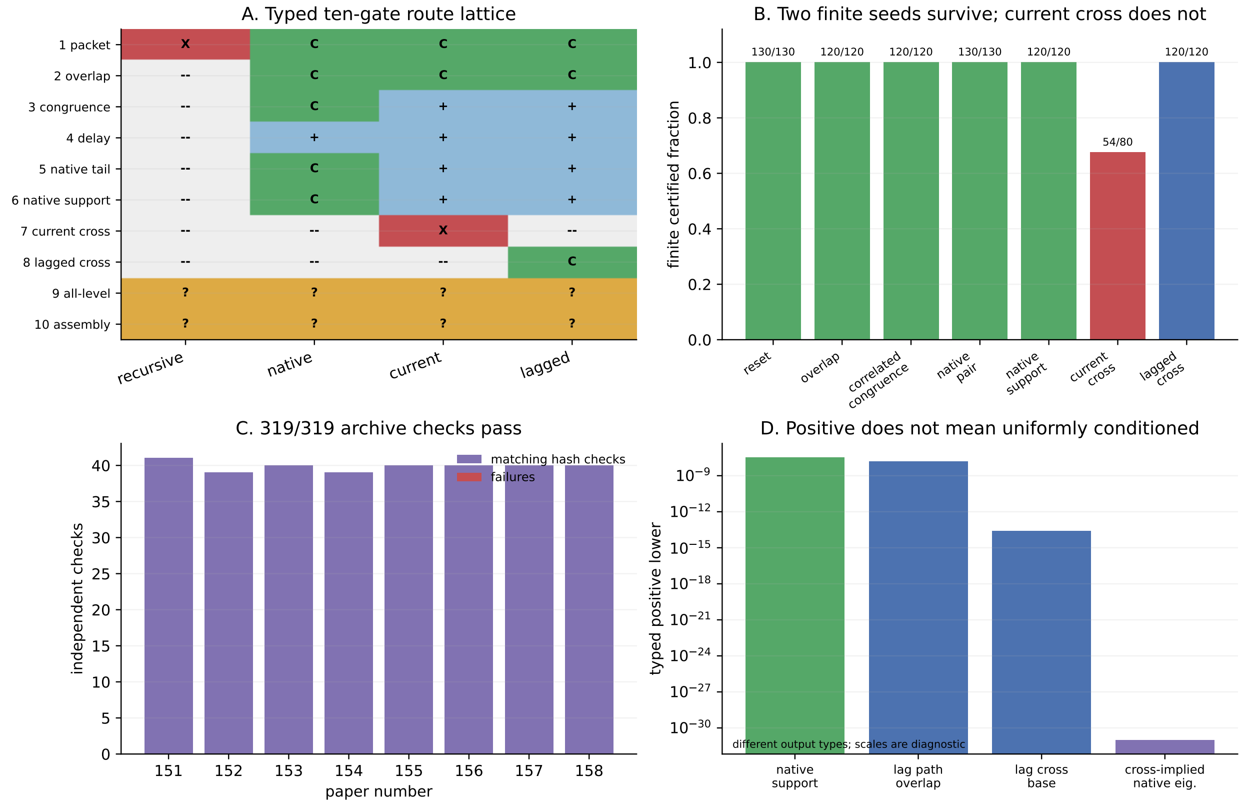


Figure 1: Typed gate matrix, finite certification counts, archive reverification, and the nonuniform scales of the surviving outputs.

The finite progression is strong but nonuniform:

$$130/130 \text{ resets}, \quad 120/120 \text{ overlaps}, \quad 130/130 \text{ native pairs}, \quad 120/120 \text{ native supports},$$

$$54/80 \text{ active current crosses}, \qquad 120/120 \text{ adaptive-lag crosses}.$$

The lagged base floor is only $2.5874 \times 10^{-14}$ and its selected path overlap floor is $1.4955 \times 10^{-8}$. Positivity is therefore settled on the finite atlas, while uniform conditioning is not.

# 6 Revised frontier

The maze has narrowed to two common gates. Gate 9 must replace five-scale counts by eventual laws for packet gaps, transition overlap, selected eigenvalue versus tail mass, spectral spread, and—only on the directional route—a bounded-lag fourth-cross lower. Gate 10 must then propagate the chosen typed seed through one consistent outward assembly. Mixing a native lower from one packet with a directional cross from another is not allowed.

The next paper should therefore state a conditional all-level theorem with two explicitly separated routes, and prove omission witnesses for each required asymptotic input. Such a theorem would be a falsifiable roadmap, not a proof that its hypotheses hold. RH-159 itself proves the block implication, the reverse nonimplication, the finite route classification, and archive integrity. It does not prove gate 9 or 10, Stage A, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

# References


Rajendra Bhatia. *Matrix Analysis*. Springer, 1997.

Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, second edition, 2013.